MIL-I-7749A(Wep)

1 June 1961

SUPERSEDING
MIL-I-7749(Aer)
20 July 1951

MILITARY SPECIFICATION

INDICATOR, TEMPERATURE, ELECTRICAL RESISTANCE, MULTIFUNCTION, -70°C TO +150°C, SINGLE

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 Scope. - This specification covers one type of single pointer, electrical resistance type, temperature indicator.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-C-5015 - Connectors, Electric, "AN" Type
MIL-I-7070 - Indicator, Temperature, Electrical Resistance, Hermetically Sealed, General Specification for
MIL-P-7936 - Parts and Equipment, Aeronautical, Preparation for Delivery

STANDARDS

FEDERAL

FED-STD-No. 595 - Colors

MILITARY

MS33639 - Cases, Instrument, Clamp-Mounted, Aircraft
MS33678 - Connectors-Electrical, Receptacles, Integral Mounting

(When requesting any of the applicable documents, refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, 5801 Tabor Avenue, Philadelphia 20 Pennsylvania, Attention: Code CDS.)

3. REQUIREMENTS

3.1 Qualification. - The indicator furnished under this specification shall be a product which has been tested, and has passed the qualifications tests specified herein, and has been listed on or approved for listing on the applicable qualified product list.

3.2 General Specification. - The requirements of Specification MIL-I-7070 apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

FSC 6685

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3.3 Construction. -

3.3.1 Case. - The outline dimensions of the case shall conform to Standard MS33639. The case size shall be 1-1/2 inch nominal. The dimension indicated as "L" shall not exceed 3 inches.

3.3.2 Scale Length. - The scale of the indicator shall be as long as practicable and shall have a minimum of 250 degrees of arc.

3.3.3 Pointer. - The pointer shall be as shown in Figure 1.

3.3.3.1 Pointer overswing. - The pointer overswing shall not be less than 3 percent of full scale angular deflection beyond each end of the scale.

3.4 Electrical Receptacle. - An electrical receptacle shall be provided and shall be located on the rear face of the case. The receptacle axis shall coincide with the intersection of the horizontal and vertical center lines of the case. The receptacle shall conform to Specification MIL-C-5015 and to Standard MS33678A10SL-3P.

The receptacle shall be wired as follows:

Terminal	Connection
A	+24V. battery.
B	Resistance bulb.
C	Common negative 24V battery and remaining resistance bulb connection.

3.5 Weight. - The weight of the complete indicator shall not exceed 1.4 pounds.

3.6 Installation. -

3.6.1 Installation Instructions. - The contractor shall furnish with each indicator one printed copy of instructions, with illustration and diagrams if necessary, covering the installation of the indicator. Previous to printing, two copies shall be furnished to the Procuring Service for approval. Instructions shall be printed on 8-1/2 x 11 inch or 17 x 11 inch paper.

3.6.2 Envelope.- An envelope furnished by the contractor containing the installation instructions shall be packaged with each instrument and shall be marked with the following information:

"IMPORTANT"

"This Envelope Contains Installation Instructions"

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification Tests. - The qualification tests of the indicator shall consist of all the tests of this specification and Specification MIL-I-7070.

4.2.1 Qualification Test Samples. - The qualification test samples shall consist of three indicators of each manufacturer's part number submitted for qualification testing. After obtaining authorization for submittal from the activity responsible for qualification, the samples shall be forwarded to the testing laboratory designated

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in the letter of authorization. Each sample shall be plainly identified by a securely attached durable tag marked with the following information: (see 6.3)

Sample for qualification tests
Submitted by (Manufacturer's name) (date) for qualification tests in accordance with the requirements of Specification MIL-I-7749 (Wep) under authorization (reference letter authorizing test)
Name of item (Same as Specification Title)
Manufacturer's Part No.
Name of manufacturer

4.3 Acceptance Tests. - Acceptance tests shall consist of all the tests of this specification and Specification MIL-I-7070.

4.4 Test Conditions. -

4.4.1 Test Voltage. - The test voltage shall be 28.5±0.5V dc, unless otherwise specified.

4.5 Test Tolerances. - The tolerances specified in table I, not enclosed in parenthesis, are applicable only when the indicators are subjected to all the tests in the sequence specified herein. The tolerances specified in table I, enclosed in parenthesis, are applicable only when the indicators are subjected to a specific test independent of sequence.

4.6 Individual Tests. -

4.6.1 Scale Error at Room Temperature. - The test resistance specified in table I shall be applied to the resistance bulb terminals of the indicator. The test voltage shall be applied and the scale errors at the test points specified in table I shall not exceed the tolerances specified in table I.

4.6.2 Position Error. - When the indicator is subjected to the Position Error Test, the maximum change in indication shall not exceed 4°C. With no voltage applied to the indicator, the pointer need not be off scale.

4.6.3 Friction Error. - When the indicator is subjected to the Friction test, the maximum difference in reading before and after tapping shall not exceed 4°C.

4.6.4 Insulation Leakage. - When the indicator is subjected to the Insulation Leakage Test, the insulation resistance shall be a minimum of 20 megohms.

4.7 Sampling Tests. -

4.7.1 Plan A. -

4.7.1.1 Reduced Voltage Operation. - The indicator shall be subjected to a Scale Error Room Temperature Test with an applied voltage of 22.5V at the test points specified in table I. The indications at the reduced voltage shall not differ from the indications with 28.5V applied by more than the tolerances specified in table I.

4.7.1.2 Magnetic Effect. - When the indicator is subjected to the Magnetic Effect test, it shall not cause the magnetic compass to deflect more than 5 degrees.

4.7.1.3 Resistance Bulb Current. - A total resistance of 100 ohms shall be applied across the bulb circuit terminals of the indicator, the test voltage applied, and the bulb circuit current measured. The current consumed in the resistance bulb circuit shall not exceed 22 ma.

4.7.1.4 Vibration Error. - When the indicator is subjected to the Vibration Error test, the maximum amplitude of pointer oscillation throughout the frequency range shall not exceed 4°C. The maximum difference between the original position of the pointer during vibration shall not exceed 4°C. Pointer oscillation shall not be considered as pointer variation.

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4.7.2 Plan B. -

4.7.2.1 Response Time. - When the indicator is subjected to the Response Time test, the response time shall not exceed 3 seconds.

4.7.2.2 Damping Factor. - When the indicator is subjected to the Damping Factor test, the damping factor shall not be less than 1.6.

4.7.2.3 Power Consumption. - When the indicator is subjected to the Power Consumption test, the power consumption shall not exceed 1.5w.

4.7.2.4 Bulb Lead Resistance Effect. - The indicator shall be connected as shown on figure 2 and subjected to a scale error at room temperature test. The indications with bulb lead resistance shall not differ from the indications without bulb lead resistance, by more than ± 3°C.

5. PREPARATION FOR DELIVERY

5.1 Packaging. -

5.1.1 Level A. - The indicator shall be packaged in accordance with Specification MIL-P-7936. The method of preservation shall be in accordance with Specification MIL-P-116 , Method 1a5, omitting preservation compound, using metal reusable containers.

5.1.2 Level C. - When this level is required, packaging shall be in accordance with standard commercial practice.

5.2 Packing. -

5.2.1 Levels A and B. - The indicator shall be packed in accordance with Specification MIL-P-7936. The level or levels required shall be as specified in the invitation for bid or contract.

5.2.2 Level C. - The indicator shall be packed to insure that the shipment arrives in a satisfactory condition at destination. The shipment shall conform to the applicable carrier's rules and regulations in effect at the time of shipment.

5.3 Marking. - The interior and exterior containers shall be marked as specified in Specification MIL-P-7936.

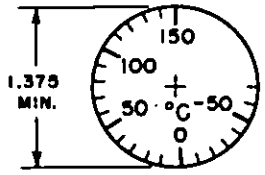
6. NOTES

6.1 Intended Use. - The electrical resistance temperature indicator covered by this specification is intended to indicate temperature in the range of -70° to 150°C, such as oil and coolant.

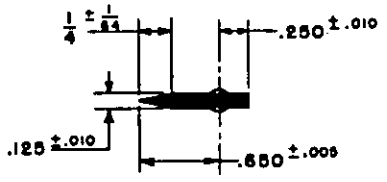
6.2 Qualification. - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Bureau of Naval Weapons, Department of the Navy, Washington 25, D. C., and information pertaining to qualification of products may be obtained from that activity.

NOTICE. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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DIAL



POINTER

MARKING	HEIGHT OR LENGTH ± 0.005	WIDTH OF LINE OR GRADUATION ± 0.005	MATERIAL OR FINISH
ALL NUMERALS MAJOR GRADUATIONS MINOR GRADUATIONS LETTERING "°C" SHADED PORTION OF POINTER	0.125 0.156 0.093 0.125 ——	0.020 0.031 0.020 0.020 ——	LUSTERLESS WHITE COLOR 37875 of FED-STD-595
BACKGROUND OF DIAL	——	——	LUSTERLESS BLACK COLOR 37038 of FED-STD-595

DIMENSIONS IN INCHES.

FIGURE 1

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TABLE I
Test Tolerances

Test Point	Test Resistance	Scale Error at Room Temperature	Reduced Voltage	Scale Error at Low Temperature
°C	Ohms	Tolerance ± °C	Tolerances ± °C	Tolerance ± °C
-70	68.27	3.0 (3.0)	2.0 (2.0)	7.0 (7.0)
-30	80.56	3.0 (3.0)	2.0 (2.0)	6.0 (6.0)
0	90.38	2.0 (2.0)	2.0 (2.0)	4.0 (4.0)
+30	100.91	3.0 (3.0)	2.0 (2.0)	6.0 (6.0)
80	120.36	3.0 (3.0)	2.0 (2.0)	6.0 (6.0)
120	137.78	3.0 (3.0)	3.0 (3.0)	6.0 (6.0)
150	151.91	3.0 (3.0)	3.0 (3.0)	7.0 (7.0)

Scale Error at High Temperature	Scale Error at Room Temperature After Open and Short Circuit	Scale Error at Room Temperature After Vibration Failure	Test Point
Tolerance ± °C	Tolerance ± °C	Tolerance ± °C	°C
7.0 (7.0)	10.0 (4.0)	10.0 (7.0)	-70
6.0 (6.0)	10.0 (4.0)	10.0 (7.0)	-30
4.0 (4.0)	10.0 (3.0)	10.0 (6.0)	0
6.0 (6.0)	10.0 (4.0)	10.0 (7.0)	+30
6.0 (6.0)	10.0 (4.0)	10.0 (7.0)	80
6.0 (6.0)	10.0 (4.0)	10.0 (7.0)	120
7.0 (7.0)	10.0 (4.0)	10.0 (7.0)	150

Test Resistance	Extreme Low Temperature Exposure		Scale Error at Room Temperature After High Temperature Exposure
	Scale Error at Low Temperature	Scale Error at Room Temperature After Low Temperature Exposure	
Ohms	Tolerance ± °C	Tolerance ± °C	Tolerance ± °C
68.27	14.0 (7.0)	10.0 (3.0)	10.0 (3.0)
80.56	13.0 (6.0)	10.0 (3.0)	10.0 (3.0)
90.38	12.0 (4.0)	10.0 (2.0)	10.0 (2.0)
100.91	13.0 (6.0)	10.0 (3.0)	10.0 (3.0)
120.36	13.0 (6.0)	10.0 (3.0)	10.0 (3.0)
137.78	13.0 (6.0)	10.0 (3.0)	10.0 (3.0)
151.91	14.0 (7.0)	10.0 (3.0)	10.0 (3.0)

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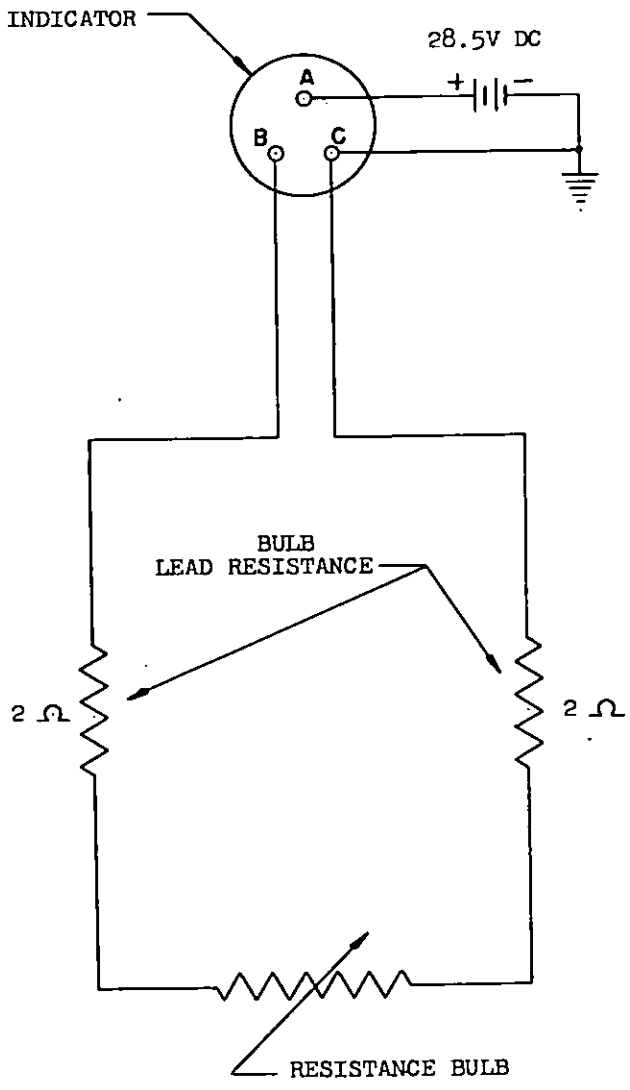


FIGURE 2. Wiring Diagram

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